DAY 34 | HANDS-ON ACTIVITIES

Observability and Monitoring: Problem Statements

Seven problems using a simulated month of production traffic with planted problems to find.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Design the Log Line

Level: Warm-up | **Concepts:** structured logging

List every field your LLM request log should contain and why. Mark which fields must be masked or never logged.

Sample run / expected result

```
request_id, ts, user_hash, release, prompt_id, model, tokens_in, tokens_out, latency_ms, retr
Never: API keys; mask: emails, phone numbers
```

Hints

- Think about what you need at 3 am during an incident.

Problem 2: Spans

Level: Easy | **Concepts:** tracing

Wrap retrieve, generate and guardrail steps in spans and print the trace for five requests. Which step dominates?

Sample run / expected result

```
trace 7f3a: retrieve 118 ms | generate 1,640 ms | guardrail 22 ms
```

Hints

- A context manager makes spans easy.

Problem 3: Monitoring Dashboard

Level: Medium | **Concepts:** metrics, plots

From the simulated month, plot daily requests, p95 latency, error rate, cost, thumbs-down rate and output tokens per request.

Sample run / expected result

```
Six small charts; thumbs-down and output tokens jump on 2026-12-18
```

Hints

- Resample by day and plot each metric separately.

Problem 4: Alert Rules

Level: Medium | **Concepts:** sustained and relative thresholds

Implement four alert rules and report which days each fired. Then tune one rule that is too noisy.

Sample run / expected result

```
slow_fixed_4s fires on 27 days (too noisy)
slow_vs_baseline fires on 12-24 only
quality_drop fires from 12-18
```

Hints

- Compare against a trailing baseline, not a fixed number only.

Problem 5: Find the Root Cause

Level: Challenging | **Concepts:** investigation

Something changed on 2026-12-18. Use the logs to find what changed (release, model, prompt) and how it affected tokens, latency and feedback.

Sample run / expected result

```
release 1.9.0 started 2026-12-18: output tokens +60%, thumbs-down rate more than doubled
```

Hints

- Group metrics by release, not just by day.

Problem 6: Topic Mix Shift

Level: Medium | **Concepts:** input drift

Compare the share of each question topic in week 1 and week 4. Which new topic appeared, and how is its "I don't know" rate?

Sample run / expected result

```
visa: 0% of questions in week 1, a growing share in week 4, "I don't know" rate 61%
```

Hints

- A new topic with many refusals usually means missing documents.

Problem 7: Feedback to Golden Set

Level: Stretch | **Concepts:** feedback loop

Build a review queue from thumbs-down answers, label five as fixed with the correct answer, and append them to the golden set with an owner and date.

Sample run / expected result

```
review queue: 212 items
added to golden_v5: 5 items (owner: mei, 2026-12-30)
```

Hints

- Each fixed failure becomes a regression test.

Ashiato (足跡): footprints